



BUTACU IONEL-CATALIN

[in linkedin.com/in/catalin-butacu](https://www.linkedin.com/in/catalin-butacu) | [✉ butacu.catalin@yahoo.com](mailto:butacu.catalin@yahoo.com)
github.com/CatalinButacu | [☎ +40 753 320 964](tel:+40753320964)

SUMMARY

C++ Software Engineer with 3+ years of professional experience developing desktop application features in Qt on a commercial industrial CAD/CAM product. Background in control systems engineering and applied informatics. Currently completing a Master's degree in Artificial Intelligence.

PROFESSIONAL EXPERIENCE

Gemini CAD Systems Software Engineer

Iași, RO
June 2023 – Present

- Built features spanning Qt's UI, networking, model/view, and event-handling layers
- Decomposed complex requirements into scoped deliverables, consistently meeting sprint estimations.
- Reduced technical debt through targeted refactoring and clean code, improving codebase resilience.
- Diagnosed and resolved production crashes and regressions through systematic root cause analysis.
- Participated in the full Scrum lifecycle: sprint planning, estimation, code reviews, sprint demos, and retrospectives.

Software Engineer Intern

August 2022 – May 2023

- Built UI components from design specifications, integrating them into existing production views.
- Delivered pixel-accurate UI and extended feature parameters through import/export integrations.
- Practised all Git workflow stages: branching, commits, merge requests, code reviews, and CI/CD.

Special Telecommunications Service (STS) IT Support Staff — Electoral IT Systems

Botoșani, RO
August 2020 – September 2020

- Provided technical support for electoral IT systems (e.g., SIMPV/SICPV).
- Assisted polling-station staff with application usage and hardware operation.

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Data Science Professional (1Z0-1110-25) [🔗](#)
- Oracle Cloud Infrastructure 2025 Generative AI Professional (1Z0-1127-25) [🔗](#)
- Oracle Cloud Infrastructure 2025 AI Vector Search Professional (1Z0-184-25) [🔗](#)
- Oracle Analytics Cloud 2025 Certified Professional (1Z0-1041-25) [🔗](#)

SKILLS & COMPETENCIES

- | | |
|--|---|
| ◦ Advanced C++ Programming (C++17/20) | ◦ Problem-Solving and Analytical Thinking |
| ◦ Qt Framework and STD Library | ◦ Technical Debt and Refactoring |
| ◦ Object-Oriented Design and Design Patterns | ◦ Memory Management, RAII, Smart Pointers |
| ◦ Multithreading and Concurrent Programming | ◦ Code Review, GitLab CI/CD, Agile/Scrum |

EDUCATION

Technical University of Iași Master of Science in Artificial Intelligence

Iași, RO
October 2024 - June 2026

Advanced Coursework: Neural Networks, Machine Learning, Natural Language Processing (NLP), Reinforcement Learning, Multi-Agent Systems, Cloud Computing, Big Data Techniques, Edge Computing Applications, Intelligent Cyber-Security

Bachelor of Science in Computer Engineering

October 2020 - June 2024

Core Curriculum: Advanced Computer Programming, Data Structures & Algorithms, Real-Time Programming, Database Management Systems, Systems Theory, Microprocessor-Based Systems, Embedded Systems Engineering, Control Engineering, Computer Networks, Neural Networks & Fuzzy Logic, Knowledge-Based Systems

PROJECTS

- Text-to-Video Generation with Physics Simulation (M.Sc. Thesis)**  **2026**
Eight-module pipeline converting natural-language prompts into physics-realistic MP4 videos. Parses text into a 3-D scene, generates assets (Shap-E), synthesises motion (SSM/KIT-ML dataset), simulates rigid-body physics (PyBullet), and optionally enhances frames with ControlNet.
- Adaptive SQL Injection Detection & Defense System**  **2026**
Team project (3 members); personal contribution: DBSCAN-based smart firewall that clusters observed attack payloads at runtime and auto-generates Snort rules, reducing ML detector load. Full system also includes a CNN classifier (95%+ accuracy) and an RL+BiLSTM adversarial attack agent.
- Edge-Deployed Hand Gesture Recognition on NPU**  **2025**
End-to-end YOLOv5n object-detection pipeline for hand gesture recognition (HaGRID dataset). Optimised for NPU/CPU deployment via TFLite INT8 quantisation, CLAHE contrast preprocessing, and temporal smoothing to reduce false positives.
- AI Medication Assistant (RAG Chatbot)**  **2025**
Full-stack RAG application: React/TypeScript frontend (GitHub Pages) backed by a Gradio API (Hugging Face Spaces). FAISS-indexed knowledge base of 1200+ medications across 20 medical categories; answers symptom queries and surfaces direct pharmacy links.
- Evolutionary Algorithms Benchmarking Framework**  **2025**
Python library implementing Canonical GA, Real-Valued GA, Differential Evolution, and Random Search. Each solver is benchmarked on standard mathematical test functions with automated CSV reporting and convergence visualisation.
- fMRI Brain-Age Prediction — 8th Place (WiDS Datathon 2025)**  **2025**
Competition ML pipeline predicting biological brain age from ~20k fMRI neuroimaging connectivity features. Gender-stratified ElasticNet model with spectral feature selection; ranked 8th nationally at the WiDS Datathon 2025 University Edition.

VOLUNTEER EXPERIENCE

- LOCAL BEST Group Iași** Iași, RO
IT Department Coordinator Sep 2023 – Aug 2024
- Directed IT operations and maintained technical infrastructure across organizational events.
 - Mentored volunteers and oversaw delivery of technical projects.
 - Administered hardware, software licensing, and network resources.
- Event Coordinator | JobShop Career Event** Nov 2022 – Apr 2023
- Coordinated logistics and operations for a large student–industry career event.
 - Managed scheduling, partner engagement, and on-site execution.
 - Improved attendee experience through strategic planning and workflow optimization.
- TOPICS Responsible | Prove It 24H Techathon** Dec 2021 – Mar 2022
- Defined technical requirements and coordinated hackathon logistics.
 - Supported participants with rules, resources, and real-time assistance.
 - Facilitated communication between organizers, industry mentors, and teams.
- IT Responsible | BEST Training Weekend** Jan 2021 – May 2022
- Developed and maintained the event website using HTML, CSS, and JavaScript.
 - Collaborated with design and marketing responsables on content, branding, and technical setup.

ADDITIONAL SKILLS

- AI / ML:** Machine Learning, Deep Learning, LLMs, RAG Systems, PyTorch, Prompt Engineering
- Web Apps:** Flask, Hadoop, React, TypeScript, Node.js, Firebase, PostgreSQL, WordPress
- Soft Skills:** Problem Decomposition, Clear Communication, Knowledge Sharing, Self-directed Learning